Voter Cluster Analysis Report Third Draft

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Introduction

LLM (Large Language Models) such as Google Gemini, Claude, ChatGPT, are increasingly deployed in domains that requires reasoning about human behavior, social dynamics, and political systems (Wikipedia, 2013). As AI has become part of everyday life in helping people draft emails, summarizing readings, answering questions, etc., these systems are also being increasingly asked to comment on political issues, including elections and voting behavior. Since democratic processes depend on informed public understanding, the way LLMs reason about and predict voting patterns is not merely a technical issue, but also a matter of civic importance. Thus, this raises an important question: **How do LLMs reason about voters and political preferences?**

Political prediction and voter modeling are central to campaigns, polling firms, advocacy organizations, and media outlets. In the real world, demographic inference models are already used to understand, estimate, and predict elections (Elder Research, 2025). Age, race, education level, income, and geographic region are often used to estimate partisan leanings. If LLMs can accurately and responsibly reason about these patterns, then they could serve as a useful educational tools or assist in comparative political analysis. However, if they rely on oversimplified stereotypes or express high confidence in uncertain predictions, then they risk reinforcing misleading narratives. Because elections are foundational to democratic societies, the way AI systems handle politically sensitive information matters.

Recent research from the paper "On the Dangers of Stochastic Parrots" argue that Large Languange Models can reproduce biases embedded in the training data and generate fluent language without true understanding (Bender, E et al., 2021). This concern is relevant to voting behavior, such as when the models are asked to predict political preference (Democrat / Republican / Libertarian / many more), from given voters' demographic information without recognizing nuance, regional variation, or change over time (Fouka & Tabellini, 2025). Thus, this is prone to stereotyping and overgeneralization.

Beyond academic research, real-world concerns about AI and elections have grown. Policymakers have warned about the role of generative AI in spreading misinformation or shaping public perception during election cycles (Yan, H et al., 2025). If a user asks an LLM, "How would this type of voter likely vote?", the answer may appear authoritative even if it is speculative, since LLMs often generate fluent and convincing responses that may nonetheless be incorrect (Huang et al., 2024). This makes it crucial to understand not only what predictions models make, but how they communicate uncertainty.

This project focuses specifically on how an LLM reasons about voting behavior under different informational conditions. Instead of only testing whether the model leans toward one political party, the study examines how it handles varying levels of detail and ambiguity. The prompts are designed to examine how the model predicts voting preferences based on demographic information, changes its prediction when information is limited, gives consistent answers to the same prompt, and whether a single response is accurate and well-calibrated.

Rather than simply asking whether the model is "correct" in a polling sense, this study focuses on the model's accuracy and consistency, as well as determining which variables (voters' information) most influence

Voter ID	County	Age Group	Gender
004	Yakima	18-24	Male
005	Snohomish	55-64	Male
006	King	55-64	Female
007	Pierce	55-64	Male
008	Pierce	35-44	Female
009	Snohomish	35-44	Unknown/Other

Table 1: Sample of simulated voter profiles

its predictions.

As LLMs become more integrated into how people access political information, it is important to understand what they predict, how consistently they do so, and what factors drive those predictions. This study aims to provide insight into the reliability and underlying reasoning patterns of LLMs in the context of voting behavior.

Experimental Design

For this experiment we have two main research questions:

1. How accurate are the predictions made by the LLM?
2. How does the consistency of predictions change when changing both the covariate levels and the covariates provided?

Research Question 1: Accuracy

To test the accuracy of the predictions, we use data from the Washington Secretary of State office (Reports, Data, and Statistics — WA Secretary of State, 2024). Their government website contains publicly available data related to voter registration, election results, and voter demographics for every year dating back to 2007.

The first step of using this data was to collect demographic information about real Washington voters and use it to create simulated Washington voters. In particular, we used the *Voter Demographics* tables from the *Voter Registration* data. Among this data, we focused on the covariates **County, Age, and Gender**.

To simulate $N = 100$ voters, (1) sample a county for each voter, using the proportion of voters in each county as probability weights, (2) use the two-way tables from the *County and Age group* and *County and Gender* data sets to calculate the conditional probabilities $P(\text{age group}|\text{county})$ and $P(\text{gender}|\text{county})$, (3) sample an age group and gender for each voter, using the above conditional probabilities as probability weights. The resulting data frame from this simulation will provide us with our simulated Washington voters data. A sample of this data can be seen in table 1. This data will be given to each question to generate responses. The first group of questions relate to party voting habits:

- What party is this voter likely to vote for in the next U.S. presidential election?
- Is this voter likely to vote democrat in the next U.S. presidential election?
- Is this voter likely to vote republican in the next U.S. presidential election?
- Is this voter likely to vote third party in the next U.S. presidential election?

The variation in these questions will provide insight into how the LLM responds when the wording of the question varies. These questions about which party each voter will vote for will provide an understanding of how the LLM interprets general voting behavior based on the given characteristics.

The second group of questions relates to specific items on a ballot. Each question was taken from the Washington 2024 General Election ballot (2024 General Election Voters' Guide — WA Secretary of State, 2024):

1. Federal Candidates
   (a) President/Vice President
   (b) Maria Cantwell or Dr Raul Garcia for U.S. Senator for Washington State
2. Statewide Candidates
   (a) Sal Mungia or Dave Larson for Supreme Court Justice Position #02
   (b) Bob Ferguson or Dave Reichert for Governor
   (c) Mike Pellicciotti or Sharon Hanek for State Treasurer
   (d) Pete Serrano or Nick Brown for Attorney General
3. Measures
   (a) Initiative Measure No. 2066 (concerns regulating energy services)
   (b) Initiative Measure No. 2109 (concerns taxes)
   (c) Initiative Measure No. 2117 (concerns carbon tax credit trading)
   (d) Initiative Measure No. 2124 (concerns state long term care insurance)

These questions will provide insight into how the LLM predicts specific voting behavior, with regard to federal and state voting, based on the given characteristics.

Given the simulated voter data, we then prompted the Google Gemini *gemma-3-27b-it* model to predict how each voter would respond to each question in both groups of questions.

Research Question 2: Consistency

To test the consistency of the predictions, we simulate voters with a wider range of covariates than we used to test accuracy. In particular the covariates we use race, age, gender, urbanicity, marital status, number of kids, religion, education, and income as covariates (for specific levels, see appendix). Levels of each covariate were sampled uniformly to simulate $N = 100$ voters. Voters were also assigned a voter ID. We then grouped the covariates into **demographics** (race, age, gender, urbanicity), **family** (marital status, number of kids, religion), and **socioeconomic status** (education, income). For each group we created a sub-data frame containing only the covariates for the respective group and the voter ID. Each data frame will then be given to each question to generate responses.

For consistency testing we add a group of questions taken from a political typology quiz from the Pew

Research Center in addition to those used for accuracy testing (Pew Research Center, 2021):

- Would this voter rather have small government providing fewer services or bigger government providing more services?
- Does this voter think 'Business corporations make too much profit' or 'Most corporations make a fair and reasonable amount of profit'
- Which comes closer to your view of candidates for political office, even if neither is exactly right? There is at least one candidate who shares most of my views. None of the candidates represent my views well.
- Which of the following statements comes closest to your view? Religion should be kept separate from government policies Government policies should support religious values and beliefs

These questions provide a more comprehensive view of political ideology than those used in the accuracy testing. We are able to use such questions as we are not limited to questions in which we have real data for. We also ask, "Is the behavior easy to predict?" for each voter. This question will give us a sense of how much the model believes it understands the voters.

Using the same Google Gemini *gemma-3-27b-it* model as before, we provide the voter data each question as a prompt to the model. We then have the model predict a response for the entire data set five times, resulting in 5 predictions per voter, per question.

Input and Output

To format the prompts, we first introduce the form of the voter data and provide the data as one long string. There is evidence showing that the order of inputs has a significant impact on the performance of LLM's, and randomizing input order can help minimize the effects of positioning bias (Rafi et al., 2024). As such, for each prompt we randomize the position that voters are presented as well as the order of the covariates. We use a voter id to track the predictions of each voter across multiple prompts.

For example, consider a reduced version of the accuracy data frame which contains the covariates county, age, and gender for 3 voters. For one prompt, the input data may be provided to the model as **"Voter data (voter id, county, age, gender): 1, King, 55-64, Male; 2, Kitsap, 25-34, Female; 3, Thurston, 45-54, Male."**. The next prompt provided to the model may then be ordered as **"Voter data (gender, age, voter_id, county): Male, 25-24, 2, Kistap; Female, 45-54, 3, Thurston; Male, 55-64, 2, King."**.

We then append the question and response options to the voter data as, **"For each voter in the voter data, *question*? Respond with only *response options*."**. We include the response options in the prompt to ensure reasonable predictions are made. For example, when asked what presidential candidate a voter would vote for in the 2024 general election, the model predicts Biden, Trump, or unknown for all voters. However, we know Biden dropped out of the presidential race and the democrat nominee to replace him was Harris. Thus, we require the model to instead choose between Harris or Trump for proper analysis. Restricting response options also prevents the model from explaining it's predictions, thus ensuring the data is in a usable format.

Finally, we append rules to the prompt. These rules were refined to ensure the output is reflecting the voter data, tracking the appropriate voter_id, and formatting the output as desired. For the accuracy prompts, the rules are **"Can only return unknown sparingly, do not omit any voter, produce EXACTLY 100 voters with voter_id's 1 to 100, output MUST be sorted by voter_id in order given, do not fabricate voter_id values only use those provide, output format: voter_id. prediction"**. For the consistency prompts the rules are **"Rules (MUST follow all): Can only return unknown sparingly, do not omit any voter, produce EXACTLY 100 predictions, return only the 100 prediction lines with no intro or explanation, for line k, use voter_ID from**

	vote_republican		
vote_democrat	no	unknown	yes
no	6	0	44
unknown	0	1	0
yes	26	0	23

Table 2: Contingency table for vote_democrat and vote_republican

row k in the voter data above, do NOT sort, group, or reorder by voter_ID; keep the exact row sequence as provided, do not fabricate voter_ID values, Format each line exactly: voter_ID. prediction.". Note that for both research questions the rules are similar but presented with different wording, each one specifically refined. While both research questions use very similar prompts, the rules are not interchangeable, as doing so results in output errors such as missing voters or incorrect ordering of predictions.

Example of prompt for consistency research question, reduced to 5 voters:

Voter data (race, voter_ID, urbanicity, age, gender): White, 1, urban, 18-24, unspecified; Asian, 39, rural, 55-64, unspecified; Native American/Alaska Native, 75, rural, 65+, female; Prefer not to say, 5, urban, 55-64, male; Multiracial, 14, rural, 18-24, nonbinary. For each voter in the voter data, how would the voter vote on the following: Initiative Measure No. 2117 concerns carbon tax credit trading. This measure would prohibit state agencies from imposing any type of carbon tax credit trading, and repeal legislation establishing a cap and invest program to reduce greenhouse gas emissions. This measure would decrease funding for investments in transportation, clean air, renewable energy, conservation, and emissions-reduction. Should this measure be enacted into law? Respond with one word, yes no. Rules (MUST follow all): Can only return unknown sparingly, do not omit any voter, Produce EXACTLY 100 predictions, Return only the 100 prediction lines with no intro or explanation, For line k, use voter_id from row k in the voter data above, Do NOT sort, group, or reorder by voter_id; keep the exact row sequence as provided, do not fabricate voter_id values, Format each line exactly: voter_id. prediction

Example of reduced output for above prompt:

1. no
39. no
75. no
5. yes
14. no

Experiment Analysis

Accuracy Analysis

Exploratory Data Analysis

Our preliminary findings indicate a lack of accuracy in voter prediction due to a lack of logical consistency across predictions. Consider the responses to the prompts (1) "What party is this voter likely to vote for in the next U.S. presidential election", (2) "Is this voter likely to vote democrat in the next U.S. presidential election?", and (3) "Is this voter likely to vote republican in the next U.S. presidential election?" - indicated as what_party, vote_democrat, and vote_republican respectively. In tables 2, 3, and 4 we see the contingency tables containing the pairwise count of voter predictions for the three questions. In table 2, we would expect most observations

	vote_democrat		
what_party	no	unknown	yes
Democrat	20	1	26
Republican	30	0	23

Table 3: Contingency table for what_party and vote_democrat

	vote_republican		
what_party	no	unknown	yes
Democrat	21	1	25
Republican	11	0	42

Table 4: Contingency table for what_party and vote_republican

to be the combinations *no and yes*, *yes and no*, and *no and no*. In table 3 we would expect most observations to be *Democrat and yes* and *Republican and no*. Similarly for table 4 we would expect most observations for *Republican and yes* and *Democrat and no*. Ideally all other prediction combinations would be low (not zero, allowing room for third party predictions) in order for predictions across prompts to maintain coherence. As we can see in the tables, this is not the case. In table 2 we observe 23 counts of *yes and yes*, suggesting 23% of simulated voters will vote both democrat and which is not a realistic option in the context of the questions. Table 3 relating to vote_democrat shows 20 voters predicted *Democrat and no*, and 23 voters predicted *Republican and yes*. In table 4 relating to vote_republican we observe 11 voters predicted *Republican and no*, and 25 voters predicted *Democrat and yes*. Again these two tables have 43% and 35% of voters with contextually illogical predictions. This begins to indicate a lack of confidence in prediction accuracy.

In Figure 1 we can see for each initiative, how many voters in each age group were predicted to vote "yes", "no", or "unknown". We know that the energy initiative passed, and the other initiatives did not pass. The results show the percentage of votes in favor of each initiative as 51.17%, 35.89%, 38.05%, and 44.54% (Washington Secretary of State, 2024). The predictions for the energy initiative are the most accurate as they reflect the small margin by which the initiative passed. The predictions for the taxes initiative are exactly evenly distributed between yes and no, which is somewhat inaccurate to the true results. The carbon tax initiative is not reflective of the results, with the overwhelming 98% of voters predicted as no, when the initiative only received 38.05% of votes as no. The insurance initiative is incorrectly predicting the initiative will pass by a more significant margin than by which the initiative was rejected. Overall, these observations also indicate a lack of confidence in prediction accuracy.

Assessing the unknowns, we see that throughout all the initiatives, only four voters were predicted as unknown. Further assessment of these voters reveals they are four distinct individuals. The two voters predicted unknown for the energy_initiative both had Unknown/Other as their gender value and were both Snohomish county. The other two voters had Female as their gender value. Two of the voters were in the age group 35-44, the other two in the 18-24 and 55-64 age groups. There was no trend between age group and gender or county. We also observe all four voters were predicted as Democrat for what_party, and all four voters had inconsistent prediction outcomes for the other party affiliation questions. Otherwise, they all received predictions for the other questions (with the exception of one voter predicted unknown for president_vote). Note that only two voters were simulated to have gender "Unknown/Other", and most of the unknown predictions in the dataset belong to one of these voters. Overall, there were only 9 instances of unknown predictions in the entire dataset, all 9 belonging to these four voters. As such, it appears the covariate combination has the most influence on the unknown predictions. However, given the small amount of unknown predictions and lack of sweeping trends,

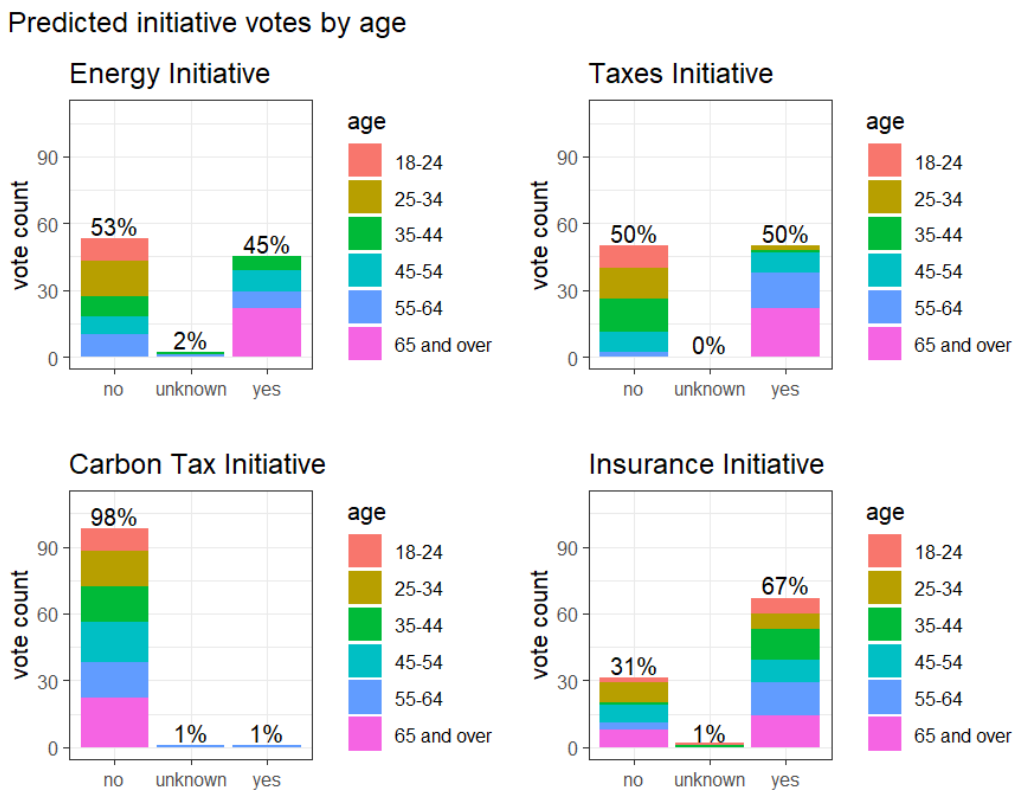


Figure 1: Initiative prediction outcomes by age

this observation is not considered concrete.

Analysis Methods

To statistically analyze the accuracy of the data, we will first conduct an exact multinomial goodness-of-fit. This test will allow us to assess the likelihood of the predicted voting behavior using real data from the Washington 2024 General Election. We use the exact multinomial test as opposed to an approximation method as the expected counts for each covariate and voter prediction combination are $<< 5$, and therefore approximation methods are unreliable. Rather than increasing the sample size of the simulated voters (which would need to be $\gtrapprox 10e5$), we use the exact multinomial test which is robust to relatively smaller sample sizes. Considering we have a large amount of observed covariate and voter prediction levels, the goodness-of-fit test is sensitive to small deviations from expected counts to observed counts. Therefore we will also aggregate over combinations of covariate levels and execute the same goodness-of-fit test on the aggregated data.

Since our sample size is 100, which is extremely small relative to the population size of Washington voters, our expected counts are going to be small. As such, the exact multionmial goodness-of-fit test will serve as a starting point to understand if the model is accurately predicting voting behavior, but we also aim to understand *how* different the prediction results are from the expected results. Therefore, we will also use Kullback–Leibler (KL) divergence to quantify the deviation (Kullback & Leibler, 1951). We will also calculate the KL divergence for the aggregated data sets for a more comprehensive understanding of the data.

For both of these tests, we first need to calculate the expected probabilities of our distribution. We make two assumptions. First, we assume that gender and age are conditionally independent given county. We make this assumption due to constraints of available data from the Washington Secretary of State office (Reports,

Data, and Statistics — WA Secretary of State, 2024). Second, we assume that party affiliation is independent of county. Since party affiliation is not required documentation under voter registration laws in Washington, there is no official data regarding party affiliation (Washington State Legislature, 2025). Therefore we must rely on third party independent estimates of party affiliation, particularly those from the Independent Voter Project (Independent Voter Project, 2025).

Given the above assumptions, to calculate the expected value for each combination of covariate levels we begin by calculating their conditional joint probability. We use the chain rule of probability to compute:

$$P(\text{age}, \text{gender}|\text{county}) = P(\text{age}|\text{county}) \cdot P(\text{gender}|\text{county})$$

We then calculate the expected value as:

$$\mathbb{E}[\text{voter prediction}, \text{county}, \text{age}, \text{gender}] = P(\text{age}, \text{gender}|\text{county}) \cdot \text{count}(\text{voter prediction}|\text{county}) \cdot \frac{100}{N}$$

Where count(voter prediction|county) is the number of voters in the given county who voted the same as the prediction, and $N = 5088509$ is the number of registered voters in Washington at the time of analysis (Reports, Data, and Statistics — WA Secretary of State, 2024). We multiply by $\frac{100}{N}$ to scale the expected counts relative to our sample size of 100 rather than relative to the entire state of Washington.

For example, consider a simulated voter from Clark county, aged $18 - 24$, who is Female. Suppose the model predicted this voter would vote Dave Reichert for Washington state governor in the 2024 general election. We calculate the proportion of voters in Washington state who live in Clark county, are ages $18 - 24$, and are Female to be $\approx 5.07\%$ (Reports, Data, and Statistics — WA Secretary of State, 2024). We also know that Dave Reichert received ≈ 1300 of the votes in the 2024 General Election (Washington Secretary of State, 2024). Therefore in a sample size of 100,

$$\mathbb{E}[\text{vote Ferguson, Clark County, 18-24, Female}] \approx 0.0013$$

The `multinom.test` function from the `ExactMultinom` package in R will compute the test statistic and p-values based on the probability mass function of the multinomial distribution, described by the expected probability of each combination of covariates and prediction values (Resin 2023, Resin 2025). The function will use Monte-Carlo estimation methods to compute these values as it is computationally more efficient due to the large amount of covariate and voter prediction combinations. At a significance level of $\alpha = 0.05$, a p-value < 0.05 will indicate the generated voter prediction data is significantly different from the expected distribution.

Additionally, the `KL` function from the `philentropy` package in R will compute the KL-divergence (D_{KL}) for each question. If the model is accurately predicting voting behavior, we would expect $D_{KL} \approx 0$. Larger values of D_{KL} indicate more deviance from the expected counts, and thus less accurate predictions (Kullback & Leibler, 1951).

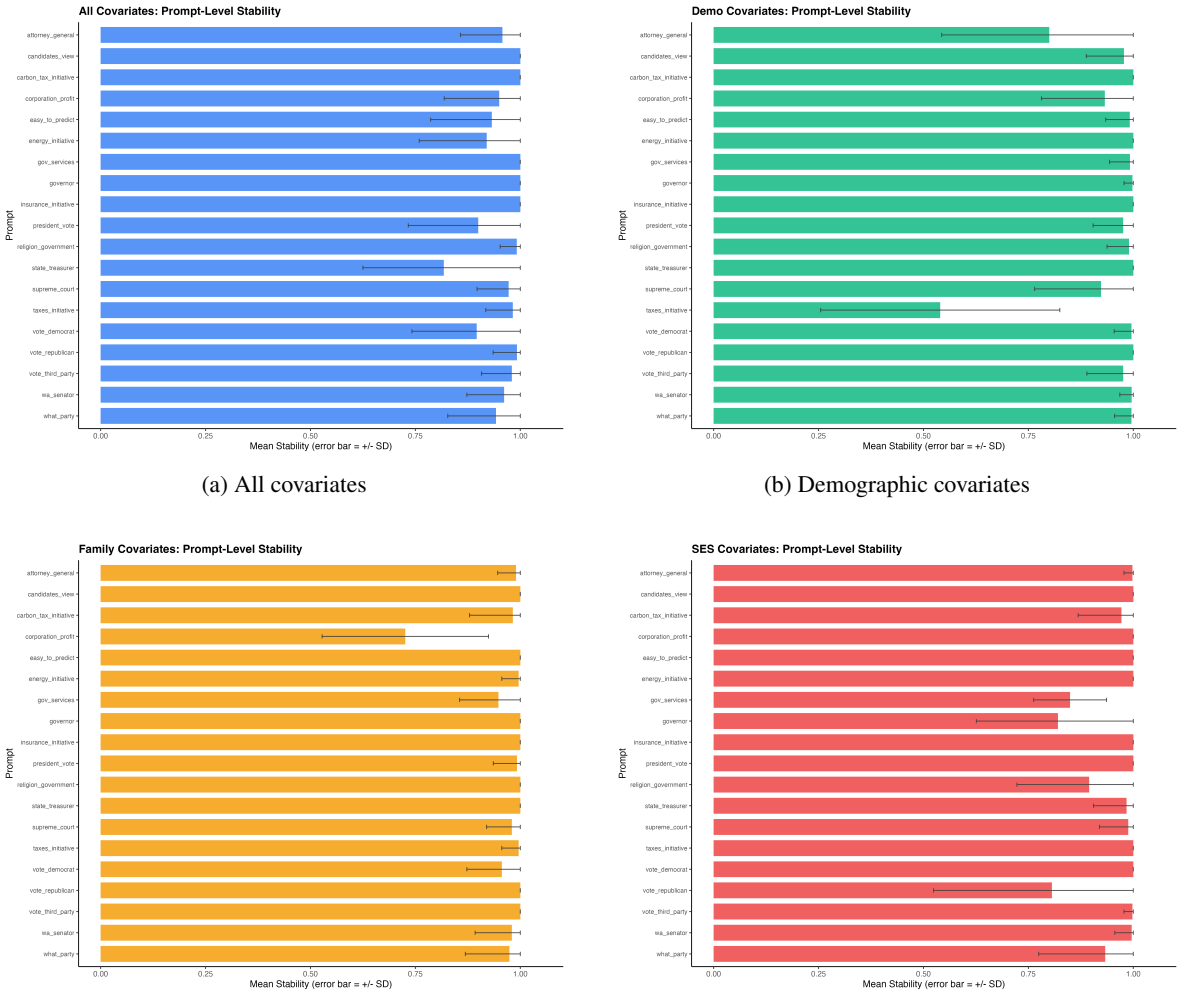
Consistency Analysis

Exploratory Data Analysis

In this section, we analyze the consistency of model output across repeated prompt variants using a stability classification method. Initially, cluster analysis was the most intuitive approach (Webster, 2020); however, due to LLM responses being a tiny discrete state space, we do not have many dimensions for response categories and no continuous variables. Therefore, after further research, stability classification is a more suitable analysis approach in this case (Bertsimas, Dunn, J., & Paskov, I. (2022)).

Before we begin the stability classification analysis, we explored and examined how stable LLM responses

are across prompts under different covariate settings, also to identify where consistency is strong versus where it fluctuates. Specifically, we explore prompt-level mean stability and variability (standard deviation) within each covariate group to detect early patterns, outliers, and potential prompt effects that may influence later inference.



(a) All covariates

(b) Demographic covariates

(c) Family covariates

(d) SES covariates

Figure 2: Predicted votes across model specifications

In the all_covariates plot in Figure 2a, stability is consistently high and tightly grouped. The overall mean is 0.9576, and it has the smallest spread across prompts (SD = 0.0481). Most prompts have short error bars, so performance is not just strong on average, but also steady from prompt to prompt. That pattern suggests the full covariate set helps the model respond in a more predictable way.

In the demo_covariates plot in Figure 2b, stability is still fairly high, but this is the weakest group overall. Its mean is 0.9518, and it has the largest variability (SD = 0.1105). Several prompts show wider error bars, which suggests demographic-only inputs are more sensitive to prompt wording. So while results are still reasonable, they are less consistent than the other setups.

In the family_covariates plot in Figure 2c, performance is the strongest overall. It has the highest mean stability (0.9747) with moderate variability (SD = 0.0622). Most prompts are near the top of the stability range, and the error bars are generally controlled. Compared with the previous run, this group now looks like the best on average rather than the most unstable.

In the ses_covariates plot in Figure 2d, stability is also strong, with mean 0.9600 and moderate spread (SD =

0.0661). Many prompts are highly stable, although variability is still higher than in all_covariates. This suggests SES variables provide useful structure for consistency, but not quite the same prompt-to-prompt uniformity as the full covariate set.

Overall, the EDA plot shows a clear split in what "best" means: family_covariates is highest on average stability, all_covariates is most uniform across prompts, and demo_covariates remains the least stable and most variable. Given that pattern, moving to stability classification and ordinal regression is the right next step to formally measure these differences and test whether they are statistically significant.

Analysis Methods

To statistically analyze the consistency of the data, we began by feeding the LLM five times under the same conditions for each prompt and stored the five responses outputs as one row in a csv file (e.g. 'attorney_general_0' to 'attorney_general_4'). We then transformed each row into a numeric stability vector by mapping each response label to a number. For instance, in 'all_covariates_attorney_general.csv', we use 'pete=1' and 'nick=0'. Thus, a row like 'pete,pete,pete,nick,nick' becomes $[1, 1, 1, 0, 0]$. This vectorization step creates a standardized representation of response patterns and allows direct comparison across rows and prompts.

Next, we computed a row-level stability score from the five responses. For binary tasks, we count how often each label appears and use the majority share:

$$\text{stability_score}_i = \frac{\max\{n_{\text{pete},i},\, n_{\text{nick},i}\}}{5}.$$

For example, in $[1, 1, 1, 0, 0]$, $n_{\text{pete}} = 3$, $n_{\text{nick}} = 2$, so

$$\text{stability_score} = \frac{3}{5} = 0.60.$$

A perfectly stable row such as 'pete,pete,pete,pete,pete' has score $5/5 = 1.00$, while a split pattern like 'pete,pete,nick,nick,nick' also has $3/5 = 0.60$. We then added this value as a new 'stability_score' column in the CSV.

After scoring each row, we assigned a 'consistency_level' column with three categories:

$$\text{consistency_level} = \begin{cases} \text{high}, & \text{if stability_score} \geq 0.75 \\ \text{medium}, & \text{if } 0.45 \leq \text{stability_score} < 0.75 \\ \text{low}, & \text{if stability_score} < 0.45. \end{cases}$$

Hence, in the attorney_general example, 'pete,pete,pete,pete,pete' is high (1.00), while 'pete,pete,pete,nick,nick' is medium (0.60). Finally, we summarized each prompt file by taking the mean and standard deviation of 'stability_score'. Conceptually, higher mean stability indicates that the response is showing a more consistent behavior, while higher standard deviation indicates greater prompt-to-prompt volatility.

At the prompt level, the same trend is visible in concrete examples. In binary-label tasks such as attorney general preference, many rows are perfectly stable (5/5 identical responses), but some rows show partial instability (3/5 versus 2/5). In multi-label tasks (for example, where responses include A, B, and unknown), instability is often driven by fragmentation across labels rather than simple flips between two categories. This indicates that uncertainty-related categories such as unknown may function as a meaningful source of inconsistency rather than random noise.

Furthermore, for deeper analysis, we also compute the overall mean and overall standard deviation across prompts to assess how different covariate information affects the stability of LLM responses.

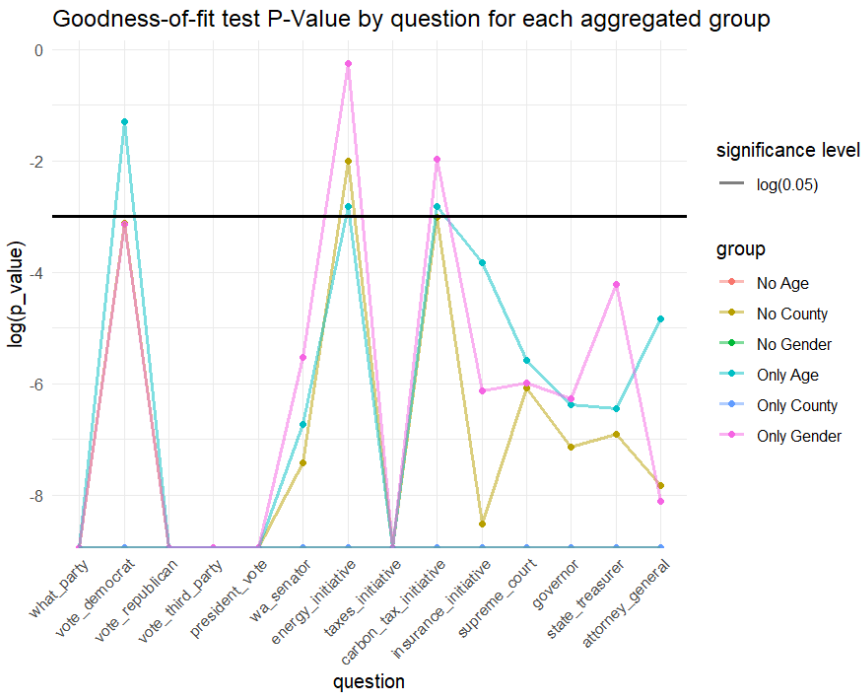


Figure 3: Goodness-of-fit p-values on aggregated data

Discussion

Accuracy Result

Using the exact multinomial test of goodness-of-fit, we find all p-values to be significant. In table 5 we see the exact multinomial goodness-of-fit test resulted in all p-values being particularly small. This indicates that the model did not accurately predict any of the questions.

When conducting the exact multinomial goodness-of-fit on the aggregated data sets we find a small quantity of p-values are not significant. In Figure 3 we can see while most p-values are significant, there are a few that are not (note that log(p-value) is being plotted for readability, and to emphasize variations among small p-values). Namely, we have no evidence to suggest the model is not accurately predicting carbon_tax_initiative voting behavior given only age and only gender; not accurately predicting energy_initiative voting behavior given age and gender, only age, and only gender; and not accurately predicting vote_democrat behavior given only age. These results indicate the model is slightly more accurate in predicting voter behavior on initiatives when not considering county. However these results are inconsistent. For example, consider the aggregate group that contains only age. The test results found the model is inaccurate when predicting vote_republican, what_party, and vote_third_party. This context indicates the lack of significance in the vote_democrat p-value does not change the overall results suggesting the model does not predict these behavioral patterns accurately.

When we look at the overall divergence results, we see some variation in D_{KL} for each question. In Figure 4 we visualize the divergence to understand the magnitude of each result. We can see that for the general questions, relating to what party the voter is most likely to vote for in the 2024 general election, D_{KL} is smaller compared to the other questions. We also see the initiative questions tend to have a slightly higher divergence, with the exception of the insurance initiative. As such, it generally appears that the model becomes more accurate for more generalized questions such as what party will a voter choose, and increasingly less accurate given specific positions or measures to vote on.

Now consider D_{KL} for the aggregated data, which can be seen in Figure 5. We see the predicted values

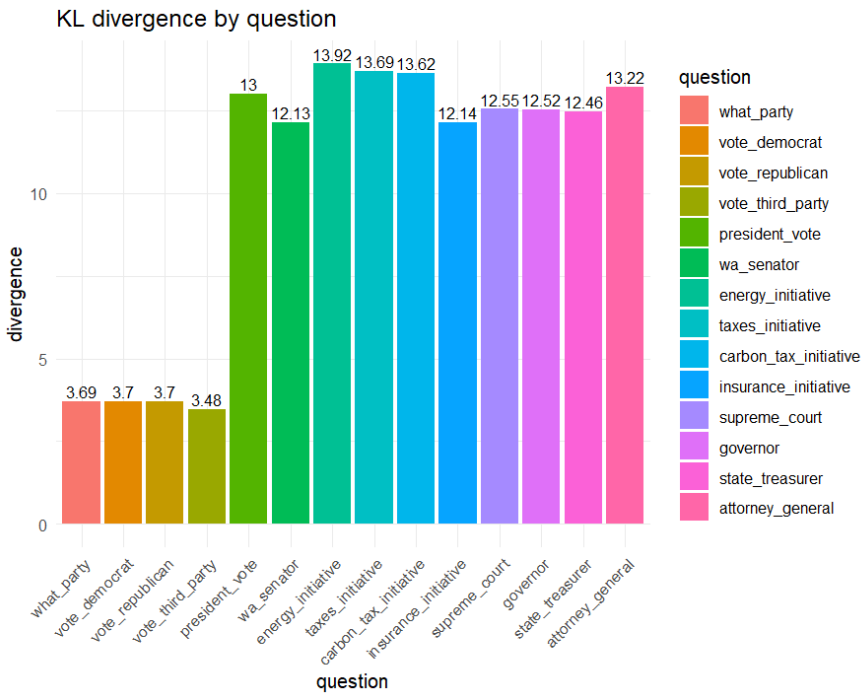


Figure 4: KL divergence plot

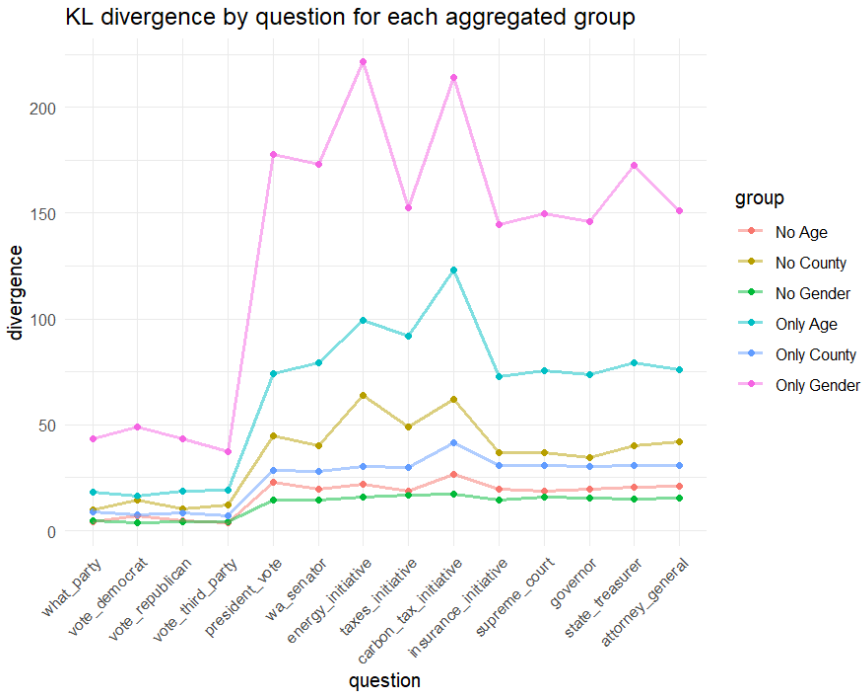


Figure 5: KL Divergence for aggregated data

tend to diverge less from the expected behavior when disregarding one covariate, or when only considering county. The model tends to diverge more when only considering age or only considering gender. We again see the same trends where the model diverges the least when predicting the general questions about party affiliation, and diverges the most for specific initiatives and specific positions. However, we see conflicting results compared to the p-value results. In the only gender, only age, and no county groups we see a large deviation for energy_initiative, but observed an insignificant p-value for the same groups and question. Similarly for the carbon_tax_initiative. Additionally, the magnitude of the deviance for these aggregated data sets is substantially higher than the deviance for the entire dataset. This provides further evidence that overall the model does not make accurate predictions.

question	p-value	D_{KL}
what_party	$<< 0$	3.691865
vote_democrat	$<< 0$	3.704098
vote_republican	$<< 0$	3.696617
vote_third_party	$<< 0$	3.477275
president_vote	$<< 0$	13.001101
wa_senator	$<< 0$	12.132791
energy_initiative	$<< 0$	13.917319
taxes_initiative	$<< 0$	13.685284
carbon_tax_initiative	$<< 0$	13.623158
insurance_initiative	$<< 0$	12.141860
supreme_court	$<< 0$	12.546537
governor	$<< 0$	12.523800
state_treasurer	$<< 0$	12.458479
attorney_general	$<< 0$	13.219101

Table 5: Accuracy goodness of fit results

Consistency Result

Covariate Group	Prompts	Overall Mean	Overall SD
all_covariates	19	0.9576	0.0481
demo_covariates	19	0.9518	0.1105
family_covariates	19	0.9747	0.0622
ses_covariates	19	0.9600	0.0661

Table 6: Stability summary results

In table 6 we have the final summary results. The final stability summary shows that all_covariates is the most consistent condition: across 19 prompts, it has the highest overall mean stability (0.9306) and the lowest variability (SD = 0.0711), compared with demo_covariates (0.8846, 0.1355), ses_covariates (0.8978, 0.1255), and family_covariates (0.8009, 0.1878) where (mean, standard deviation). This suggests that giving the LLM more complete covariate information leads to more stable and less random responses across repeated prompts.

For a further analysis, to evaluate whether all_covariates is more consistent than the others, with statistical inference and not just comparing the means (GeeksforGeeks, 2025), we fit an ordinal regression model using consistency_level (low, medium, high) as the outcome and covariate group and prompt as the predictors (consistency_level $\sim$ all_covariates + prompt), with all_covariates as the reference category to directly test whether it yields higher consistency than the other conditions. By looking at the raw data first, seen in table 7, the responses were overwhelmingly concentrated in the high consistency category across all group (proportions are all above 90%). As we can see, 91.04% for demo_covariates, 95.46% for family_covariates and 93.21% for ses_covariates.

covariate_group	consistency_level	n	prop
all_covariates	low	9	0.0048
all_covariates	medium	119	0.0636
all_covariates	high	1743	0.9316
demo_covariates	low	74	0.0394
demo_covariates	medium	94	0.0501
demo_covariates	high	1708	0.9104
family_covariates	low	6	0.0032
family_covariates	medium	80	0.0422
family_covariates	high	1810	0.9546
ses_covariates	low	34	0.0183
ses_covariates	medium	92	0.0496
ses_covariates	high	1730	0.9321

Table 7: Consistency-Level Distribution by Covariate Group

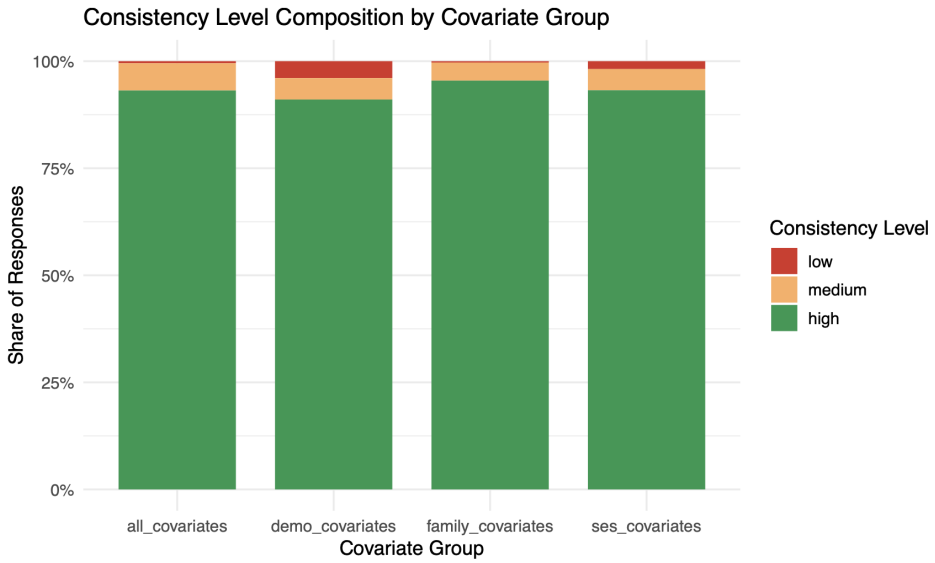


Figure 6: Consistency-Level Distribution by Covariate Group Visualization

Meanwhile the low consistency responses were rare (low proportion) across the board. Thus, these proportions suggest that all groups performed well overall. The results can be seen in Figure 6 for better readability.

However, simply comparing percentages does not account for the ordering of the outcome or control for other variables in the data. Therefore, to move beyond comparison and formally test whether the differences between groups were statistically meaningful, we performed ordinal regression below.

The results in table 8 show that the groups did not all perform equally, even after accounting for the overall high rate of consistency. Compared to all_covariates, the family_covariates group had significantly higher odds of producing a high-consistency response (OR = 1.591, 95% CI [1.193, 2.123], $p = 0.0016$), meaning that the prompts built around family-related variables were more likely to yield stable outputs. On the other end,

comparison	estimate	std_error	t_value	p_value	odds_ratio	conf_low	conf_high
demo_covariates	-0.4082	0.1265	-3.2267	0.0013	0.6649	0.5189	0.8519
family_covariates	0.4645	0.1471	3.1584	0.0016	1.5912	1.1927	2.1228
ses_covariates	-0.0115	0.1342	-0.0856	0.9318	0.9886	0.7600	1.2860

Table 8: Ordinal Regression Effects for Covariate Groups

covariate_group	consistency_level	prop
all_covariates	low	0.0260
all_covariates	medium	0.0842
all_covariates	high	0.8898
demo_covariates	low	0.0386
demo_covariates	medium	0.1185
demo_covariates	high	0.8429
family_covariates	low	0.0165
family_covariates	medium	0.0557
family_covariates	high	0.9278
ses_covariates	low	0.0263
ses_covariates	medium	0.0851
ses_covariates	high	0.8886

Table 9: Ordinal Regression Effects for Covariate Groups

demo_covariates had significantly lower odds (OR = 0.665, 95% CI [0.519, 0.852], $p = 0.0013$), suggesting that demographic covariates were associated with less consistent responses relative to the baseline. The ses_covariates group showed no meaningful difference from all_covariates (OR = 0.989, $p = 0.932$), making it statistically the closest match to the full covariate condition. While odds ratios capture the relative differences between groups, they can be harder to interpret in practical terms, so predicted probabilities were also computed below to make the results more concrete and easier to compare directly.

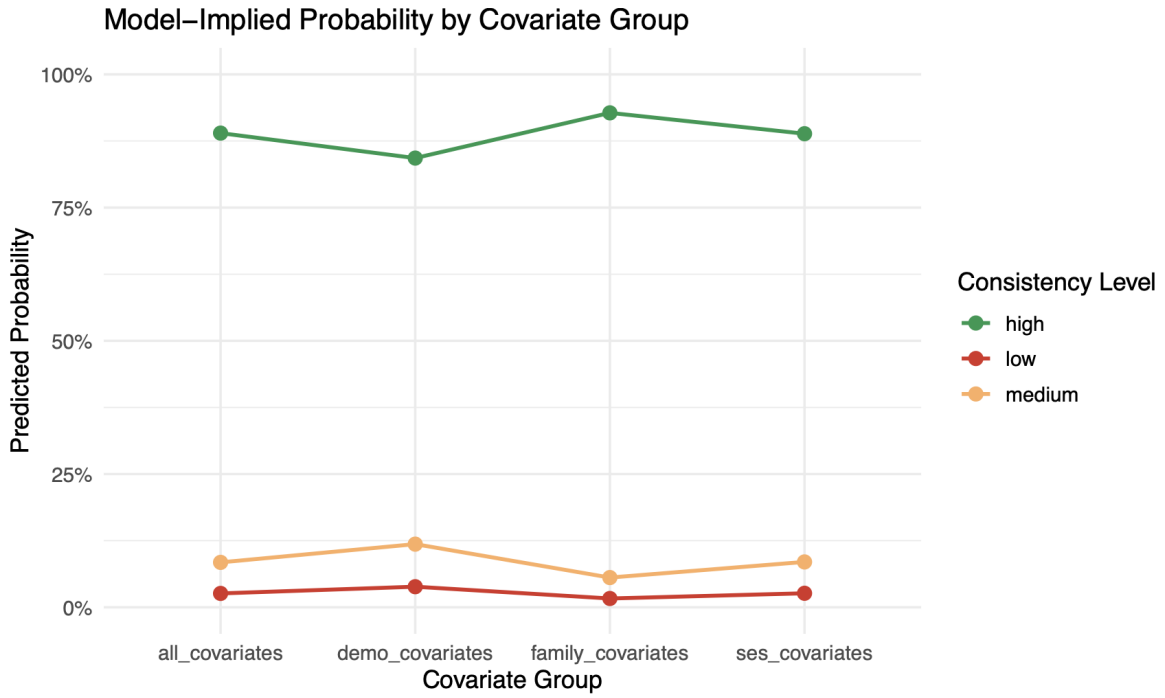


Figure 7: Model-Implied Probability by Covariate Group

As we can see in table 9, the model's predicted probabilities translated these findings into more intuitive numbers and reinforced the same pattern. Since the model also accounted for prompt type as a predictor, the predicted probabilities were calculated by fixing the prompt effect at a constant baseline. In this way, any differences in the probabilities across groups reflect the influence of covariate group alone, rather than being mixed up with which prompt was used. Under this condition, family_covariates had the highest predicted

probability of landing in the high consistency category (92.78%) and the lowest probability of low consistency (1.65%). On the other hand, demo_covariates had the lowest predicted probability of high consistency (84.29%) and the highest probability of low consistency (3.86%). The all_covariates and ses_covariates groups fell in between and were nearly identical to each other (high consistency: 88.98% vs. 88.86%), which aligns with their non-significant model contrast and suggests that using SES covariates (education, income) alone produces consistency levels comparable to using all covariates together. These results can be seen in Figure 7 for better readability.

Conclusions

The results of the accuracy analysis indicate that with the given covariates of age, gender, and county, the LLM is not able to predict voter behavior with a high degree of accuracy. The model tends to be more reliable for general questions, such as party affiliation, and less reliable for precise questions, such as voting on specific positions. However we emphasize that these more reliable results are accompanied by inconsistent and illogical behavior predictions, essentially negating any stability present. Additionally, as we saw in KL divergence by aggregated groups, the data set not considering age had the highest divergence of all. Overall this reveals an over-reliance on specific covariates.

It is important to note that this analysis of accuracy is limited by available data. As discussed in the introduction, there are many covariates beyond those we utilized such as race, education level, income, and more. This analysis is also limited specifically to a general election within Washington State, in the U.S.. Future work can expand the accuracy analysis not only to other states in the U.S. but to other countries, and other types of elections as well. Given the amount of data available for other groups of people (and thus potentially more data for the model to be trained on), the results may change.

Generally, the accuracy results suggest the LLM is able to understand that voting behaviors are influenced by factors such as location, age, or gender. While this analysis does not recommend LLM's to be used as proxies for forecasting election results, there appears to be room for further model refinement in which case they may be used as explantory tools for political behavior.

All in all, these findings suggest that while the model generally performed well across all conditions, the type of covariates included in the prompt does have a measurable effect on response consistency. Interestingly, family_covariates outperformed all_covariates, which hints that adding demographic or SES information alongside family variables may introduce some noise that slightly reduces consistency. The weaker performance of demo_covariates is also particularly noteworthy, as it may indicate that the model is more sensitive o variable when demographic information is included in the prompt, and this pattern warrants further investigation in future work. Therefore, based on the results, family_covariates is the best-performing model since it had the highest odds of producing high-consistency responses (OR = 1.591), the highest predicted probability of high consistency (92.78%), and the lowest predicted probability of low consistency (1.65%) among all groups.

The results of the consistency analysis directly connect to the questions raised in the introduction about how LLMs reason about voters and how reliable their predictions are. First, the analysis shows that the model generally produces highly consistent predictions across repeated runs of the same prompt. This indicates that its responses are not random but instead, it follows stable internal patterns when inferring voting preferences from voter characteristics. Second, the comparison across covariate groups demonstrates that the amount and type of information provided in the prompt affects this stability. When information is limited, such as in the demographic-only prompts, the model's responses become more variable, suggesting that predictions are more sensitive to prompt wording when fewer contextual signals are available. In contrast, prompts that include richer contextual information, particularly family-related variables, lead to more consistent outputs. This suggests that the model relies on certain types of social context more strongly when reasoning about voting behavior. Finally,

while the model's high consistency indicates that it applies systematic patterns in its predictions, consistency alone does not guarantee that a single prediction is accurate or well-calibrated. Instead, the results highlight that LLMs can produce stable responses even when the underlying reasoning may rely on simplified associations between voter characteristics and political preferences, reinforcing the importance of evaluating both reliability and potential bias when using LLMs to analyze political behavior.

In addition, from the perspective of the concerns discussed earlier in introduction section about stereotyping and uncertainty in LLM political reasoning (Bender, E., McMillan-Major, A., Shmitchell, S., & Gebru, T., 2021), these results highlight an important trade-off. High consistency suggests that the model applies systematic patterns when making predictions about voters, which can make its behavior looks reliable. However, consistent predictions do not necessarily imply accurate or unbiased reasoning. If the model is drawing on simplified correlations between social characteristics and political preferences, then highly stable outputs may reflect the repeated application of those assumptions. Therefore, understanding both the consistency and the content of the model's predictions is crucial when evaluating the role of LLMs in political analysis and public discourse.

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Appendix

Question and Response Prompts

General questions

1. what_party: For each voter in the voter data, what party is the voter likely to vote for in the next U.S. presidential election? Respond with one word, name of party.

2. vote_democrat: For each voter in the voter data, is the voter likely to vote democrat in the next U.S. presidential election? Respond with one word, yes no.

3. vote_republican: For each voter in the voter data, is the voter likely to vote republican in the next U.S. presidential election? Respond with one word, yes no. item vote_third_party: For each voter in the voter data, is the voter likely to vote third party in the next U.S. presidential election? Respond with one word, yes no.

Specific Ballot Items (2024 General Election Voters' Guide — WA Secretary of State, 2024)

1. president_vote: For each voter in the voter data, who would the voter vote for in the 2024 general election for President/Vice President? Respond with only the first name of the presidential candidate, Donald or Kamala.

2. wa_senator: For each voter in the voter data, would the voter vote for Maria Cantwell or Dr Raul Garcia for U.S. Senator for Washington State in the 2024 General Election? Respond with only the first name of the candidate.

3. energy_initiative: For each voter in the voter data, how would the voter vote on the following: Initiative Measure No. 2066 concerns regulating energy services, including natural gas and electrification. This measure would repeal or prohibit certain laws and regulations that discourage natural gas use and/or promote electrification, and require certain utilities and local governments to provide natural gas to eligible customers. Should this measure be enacted into law? Respond with one word, yes no.

4. taxes_initiative: For each voter in the voter data, how would the voter vote on the following: Initiative Measure No. 2109 concerns taxes. This measure would repeal an excise tax imposed on the sale or exchange of certain long-term capital assets by individuals who have annual capital gains of over $250,000. This measure would decrease funding for K-12 education, higher education, school construction, early learning, and childcare. Should this measure be enacted into law? Respond with one word, yes no.

5. carbon_tax_initiative: For each voter in the voter data, how would the voter vote on the following: Initiative Measure No. 2117 concerns carbon tax credit trading. This measure would prohibit state agencies from imposing any type of carbon tax credit trading, and repeal legislation establishing a cap and invest program to reduce greenhouse gas emissions. This measure would decrease funding for investments in transportation, clean air, renewable energy, conservation, and emissions-reduction. Should this measure be enacted into law? Respond with one word, yes no.

6. insurance_initiative: For each voter in the voter data, how would the voter vote on the following: Initiative Measure No. 2124 concerns state long term care insurance. This measure would provide that employees and self-employed people must elect to keep coverage under RCW 50B.04 and could opt-out any time. It would also repeal a law governing an exemption for employees. This measure would decrease funding

for Washington's public insurance program providing long-term care benefits and services. Should this measure be enacted into law?Respond with one word, yes no.

7. supreme_court: For each voter in the voter data, would the voter vote for Sal Mungia or Dave Larson for Supreme Court Justice Position #02 in Washington State, in the 2024 general election?, Respond with only the first name of the candidate.

8. governor: For each voter in the voter data, would the voter vote for Bob Ferguson or Dave Reichert for Governor in Washington State, in the 2024 general election?, Respond with only the first name of the candidate.

9. state_treasurer: For each voter in the voter data, would the voter vote for Mike Pellicciotti or Sharon Hanek for State Treasurer in Washington State, in the 2024 general election?, Respond with only the first name of the candidate.

10. attorney_general: For each voter in the voter data, would the voter vote for Pete Serrano or Nick Brown for Attorney General in Washington State, in the 2024 general election?, Respond with only the first name of the candidate.

Consistency Specific Questions ((Pew Research Center, 2021)

1. easy_to_predict: For each voter in the voter data, is their behavior easy to predict? Respond with one word, yes no.

2. gov_services: For each voter in the voter data , does the voter think A: small government providing fewer services; or B: bigger government providing more services; Respond with only A B.

3. corporation_profit: Does this voter think A: Business corporations make too much profit or B: Most corporations make a fair and reasonable amount of profit.

4. candidates_view: For each voter in the voter data, which comes closer to the voters view of candidates for political office, even if neither is exactly right? A: There is at least one candidate who shares most of my views; B: None of the candidates represent my views well. Respond with only A B.

5. religion_government: For each voter in the voter data, which of the following statements comes closest to their view? Religion should be kept separate from government policies Government policies should support religious values and beliefs. Respond with only A B.

Accuracy Covariates

- County: *all 39 counties in Washington State*
- Age: 18-24; 25-34; 35-44; 45-54; 55-64; 65 and over
- Gender: Female; Male; Unknown/Other

Consistency Covariates

- Age: 18-24; 25-34; 35-44; 45-54; 55-64; 65+
- Gender: female; male; nonbinary; unspecified
- Race: White; Black/African American; Hispanic/Latino; Asian; Native American/Alaska Native; Native Hawaiian/Pacific Islander; Middle Eastern/North African; Multiracial; Other; prefer not to say
- Urbanicity: urban; suburban; rural
- Marital Status: single/never married; married; partnered; divorced; separated; widowed; prefer not to say
- Number of Kids: 0, 1, 2, 3, 4+
- Religion: none/agnostic/atheist; Christian; Jewish; Muslim; Hindu; Buddhist; Sikh; Other; prefer not to say
- Education: less than HS; HS/GED; some college; associate; bachelor; graduate/professional; prefer not to say
- Income: <$25k; $25k-$49k; $50k-$74k; $75k-$99k; $100k-$149k; $150k+; prefer not to say